



Topic 23

Mirrors and Lenses

Reading Question 23-1 (1 of 2)

What is the difference between a virtual image and a real image?

1. A virtual image will focus on a screen but doesn't exist.
2. A real image only exists in the mirror.
3. A virtual image appears to be located where no light rays intersect, whereas a real image exists where light rays actually intersect.
4. A virtual image appears to be in front of the mirror, and a real image appears to be behind the mirror.

Reading Question 23-1 (2 of 2)

What is the difference between a virtual image and a real image?

1. A virtual image will focus on a screen but doesn't exist.
2. A real image only exists in the mirror.
3. **Answer: A virtual image appears to be located where no light rays intersect, whereas a real image exists where light rays actually intersect.**
4. A virtual image appears to be in front of the mirror, and a real image appears to be behind the mirror.

Reading Question 23-2 (1 of 2)

The image created by a flat mirror alone is

1. virtual, with a magnification $M = 1$.
2. real, with a magnification $M = 1$.
3. virtual, with some magnification greater or less than 1.
4. real, with some magnification greater or less than 1.

Reading Question 23-2 (2 of 2)

The image created by a flat mirror alone is

1. **Answer: virtual, with a magnification $M = 1$.**
2. real, with a magnification $M = 1$.
3. virtual, with some magnification greater or less than 1.
4. real, with some magnification greater or less than 1.

Reading Question 23-3 (1 of 2)

A mirror creates an image of an object; it is upright, and appears to be slightly smaller than the object itself. What kind of mirror is it?

1. concave
2. convex
3. flat
4. unknown—not enough information given

Reading Question 23-3 (2 of 2)

A mirror creates an image of an object; it is upright, and appears to be slightly smaller than the object itself. What kind of mirror is it?

1. concave
2. **Answer: convex**
3. flat
4. unknown—not enough information given

Reading Question 23-4 (1 of 2)

When the Sun is setting with its bottom edge just touching the horizon,

1. the Sun is actually where it appears to be.
2. the Sun is just below the horizon.
3. the Sun is about one diameter above the horizon.
4. the Sun is 2.9° above the horizon

Reading Question 23-4 (2 of 2)

When the Sun is setting with its bottom edge just touching the horizon,

1. the Sun is actually where it appears to be.
- 2. Answer: the Sun is just below the horizon.**
3. the Sun is about one diameter above the horizon.
4. the Sun is 2.9° above the horizon

Reading Question 23-5 (1 of 2)

Which of the following do not suffer from chromatic aberration?

1. convex lenses
2. concave lenses
3. concave mirrors
4. none of these choices

Reading Question 23-5 (2 of 2)

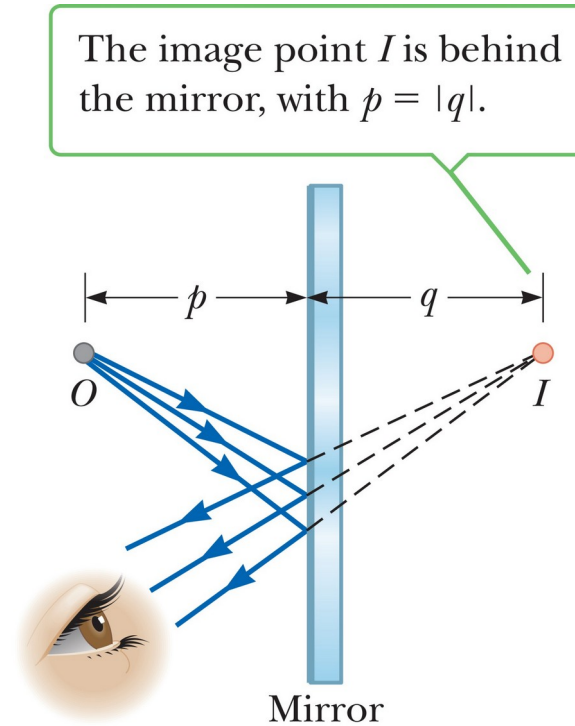
Which of the following do not suffer from chromatic aberration?

1. convex lenses
2. concave lenses
3. **Answer: concave mirrors**
4. none of these choices

Topic 23: Mirrors and Lenses

TOPIC DISCUSSION

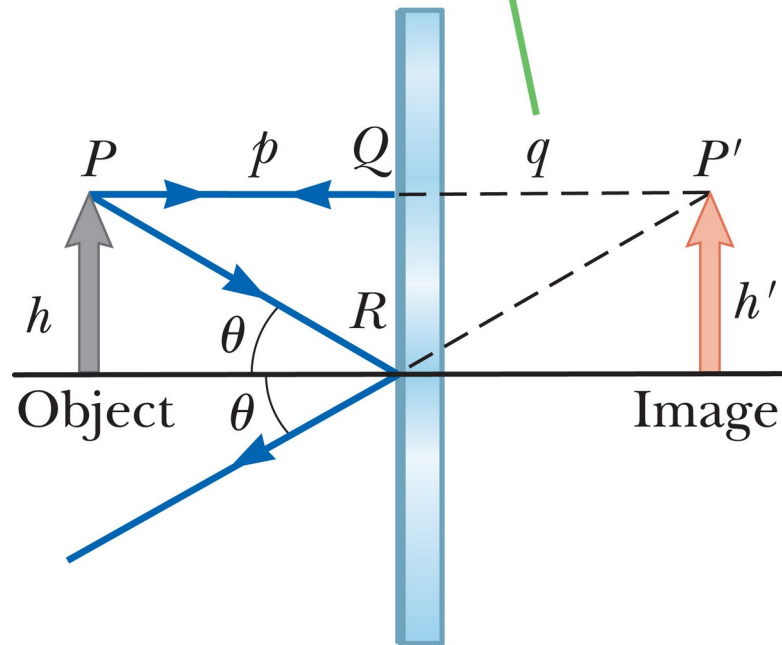
Flat Mirrors (1 of 4)



Images are formed at the point where rays of light actually intersect or where they appear to originate.

Flat Mirrors (2 of 4)

Because the triangles PQR and $P'QR$ are identical,
 $p = |q|$ and $h = h'$.



$$M \equiv \frac{\text{image height}}{\text{object height}} = \frac{h'}{h}$$

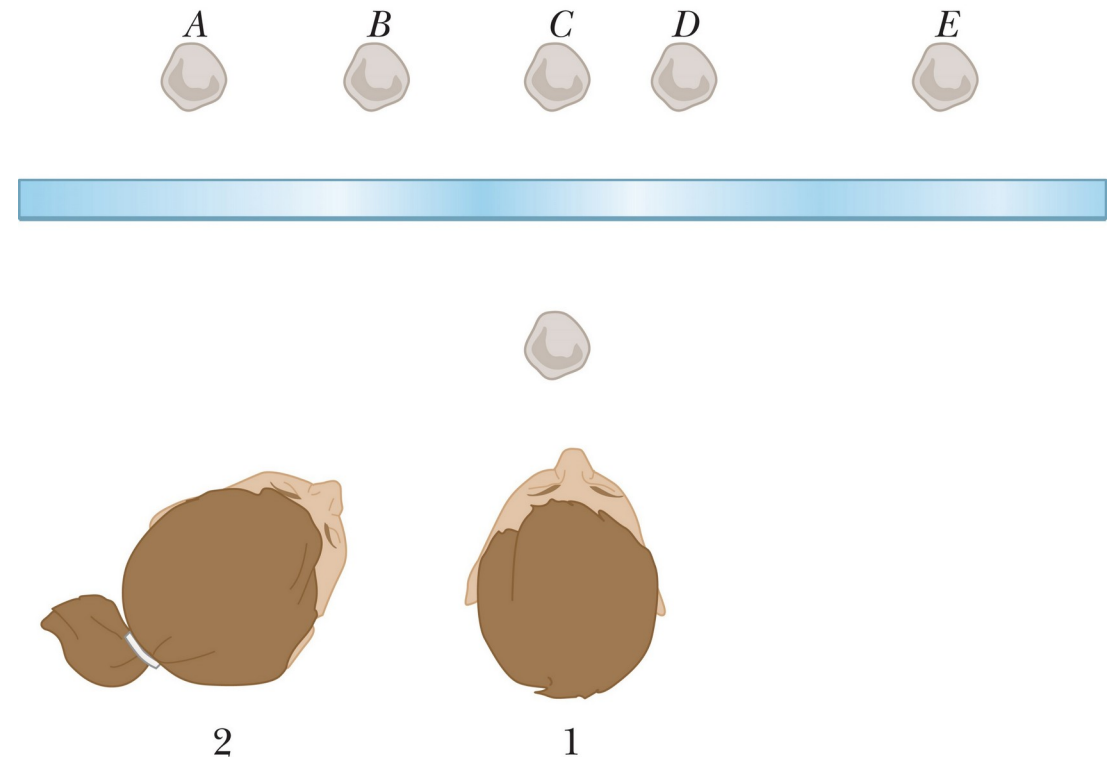
Flat Mirrors (3 of 4)

1. The image is as far behind the mirror as the object is in front.
2. The image is unmagnified, virtual, and upright.

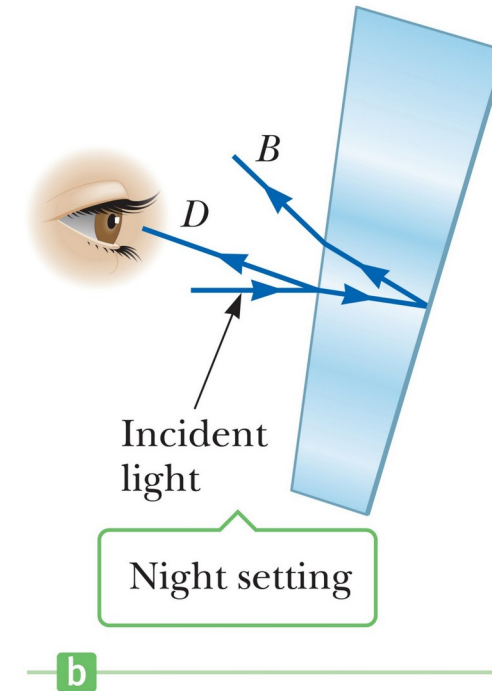
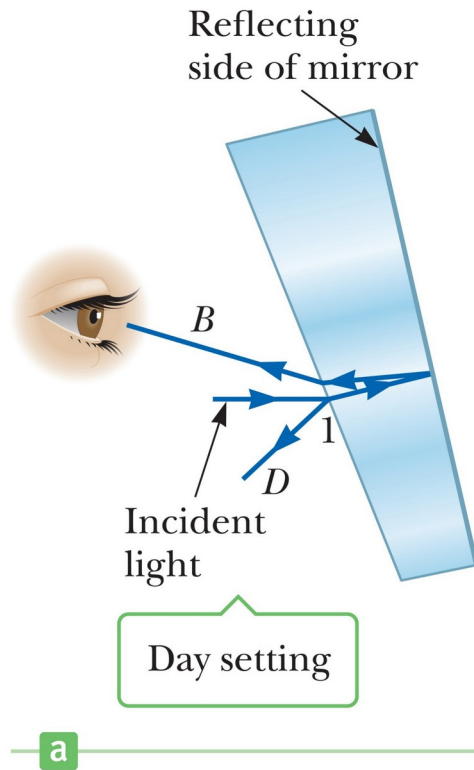
Think – Pair – Share 1

In the overhead view of the figure below, the image of the stone seen by observer 1 is at C. Where does observer 2 see the image, at A, at B, at C, at D, at E, or not at all?

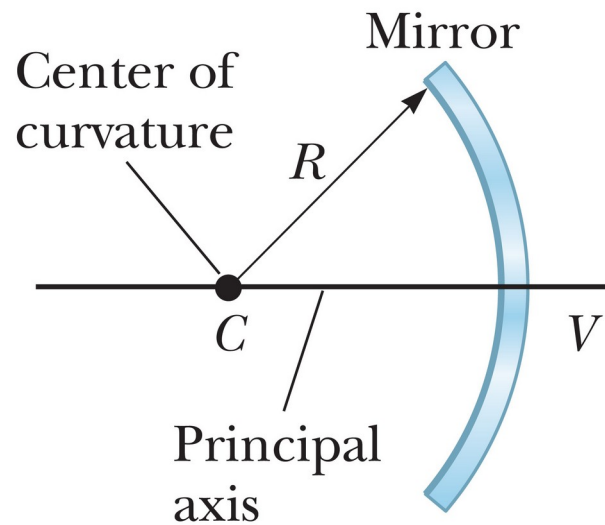
1. at point A
2. at point B
3. at point C
4. at point D
5. at point E



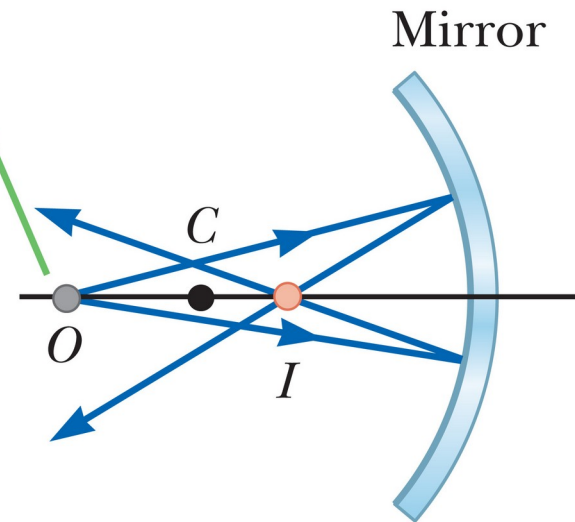
Flat Mirrors (4 of 4)



Images Formed by Spherical Mirrors (1 of 4)

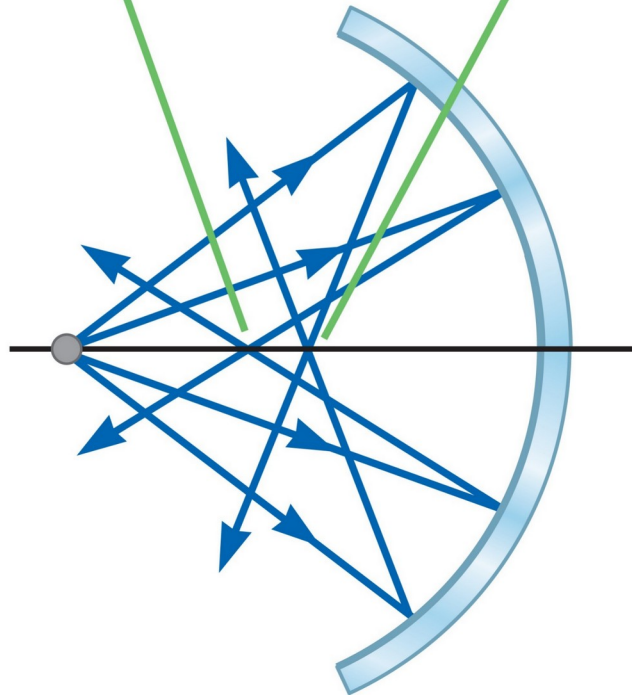


If the rays diverge from O at small angles, they all reflect through the same image point at I .

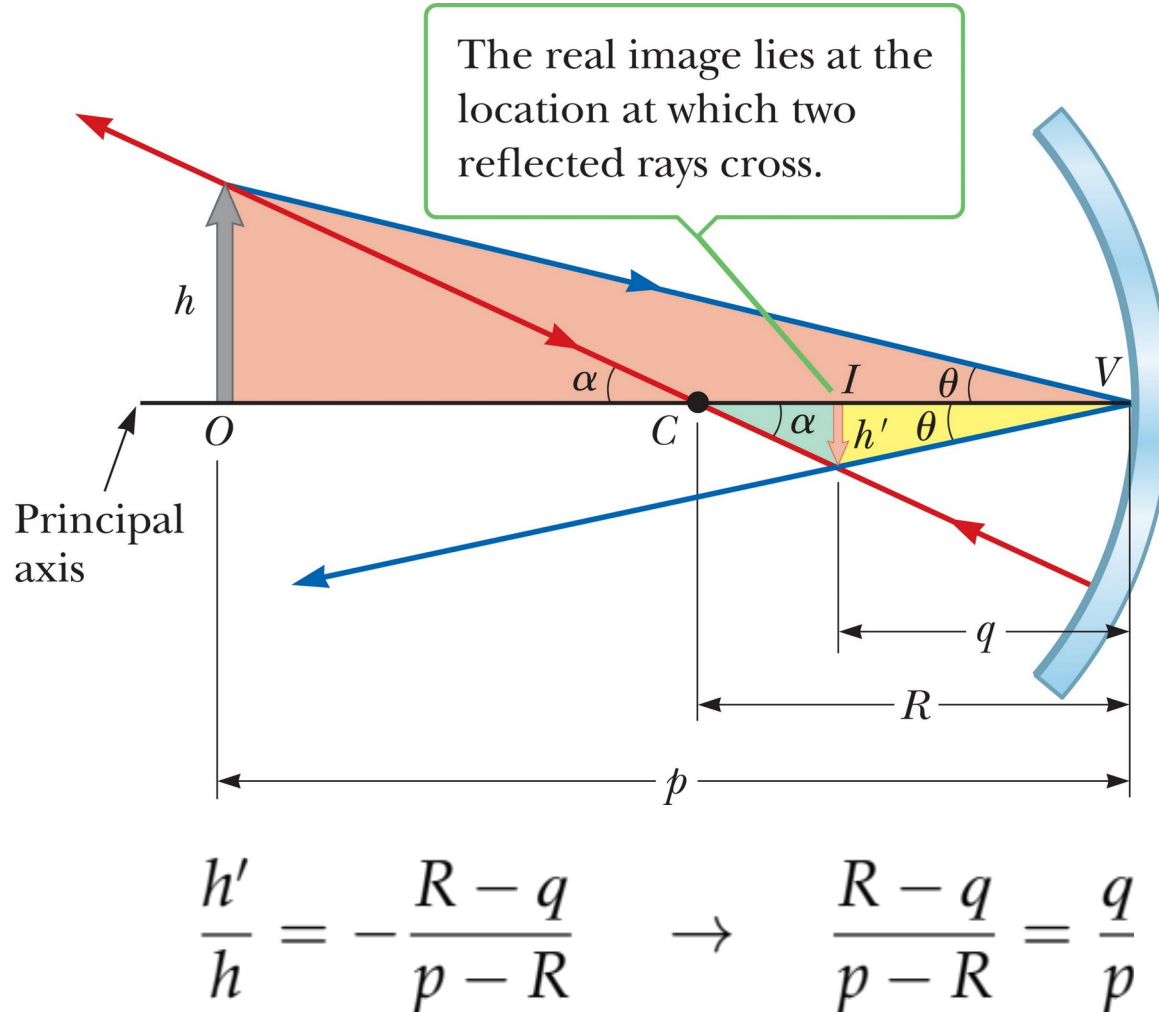


Images Formed by Spherical Mirrors (2 of 4)

The reflected rays intersect at different points on the principal axis.



Images Formed by Spherical Mirrors (3 of 4)



$$M = \frac{h'}{h} = -\frac{q}{p}$$

$$\tan \alpha = \frac{h}{p - R}$$

$$\tan \alpha = -\frac{h'}{R - q}$$

$$\frac{1}{p} + \frac{1}{q} = \frac{2}{R}$$

Images Formed by Spherical Mirrors (4 of 4)

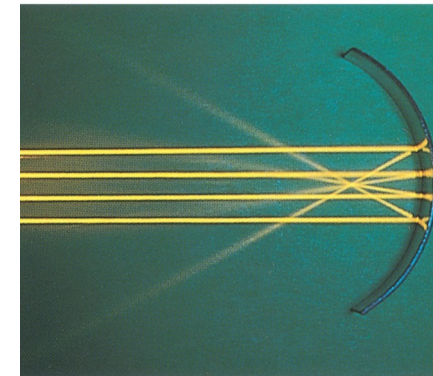
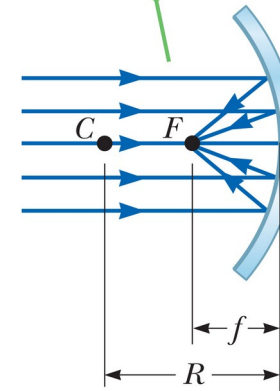
$$p \rightarrow \infty \text{ then } \frac{1}{p} \approx 0$$

$$\frac{1}{p} + \frac{1}{q} = \frac{2}{R} \rightarrow q \approx \frac{R}{2}$$

$$f = \frac{R}{2}$$

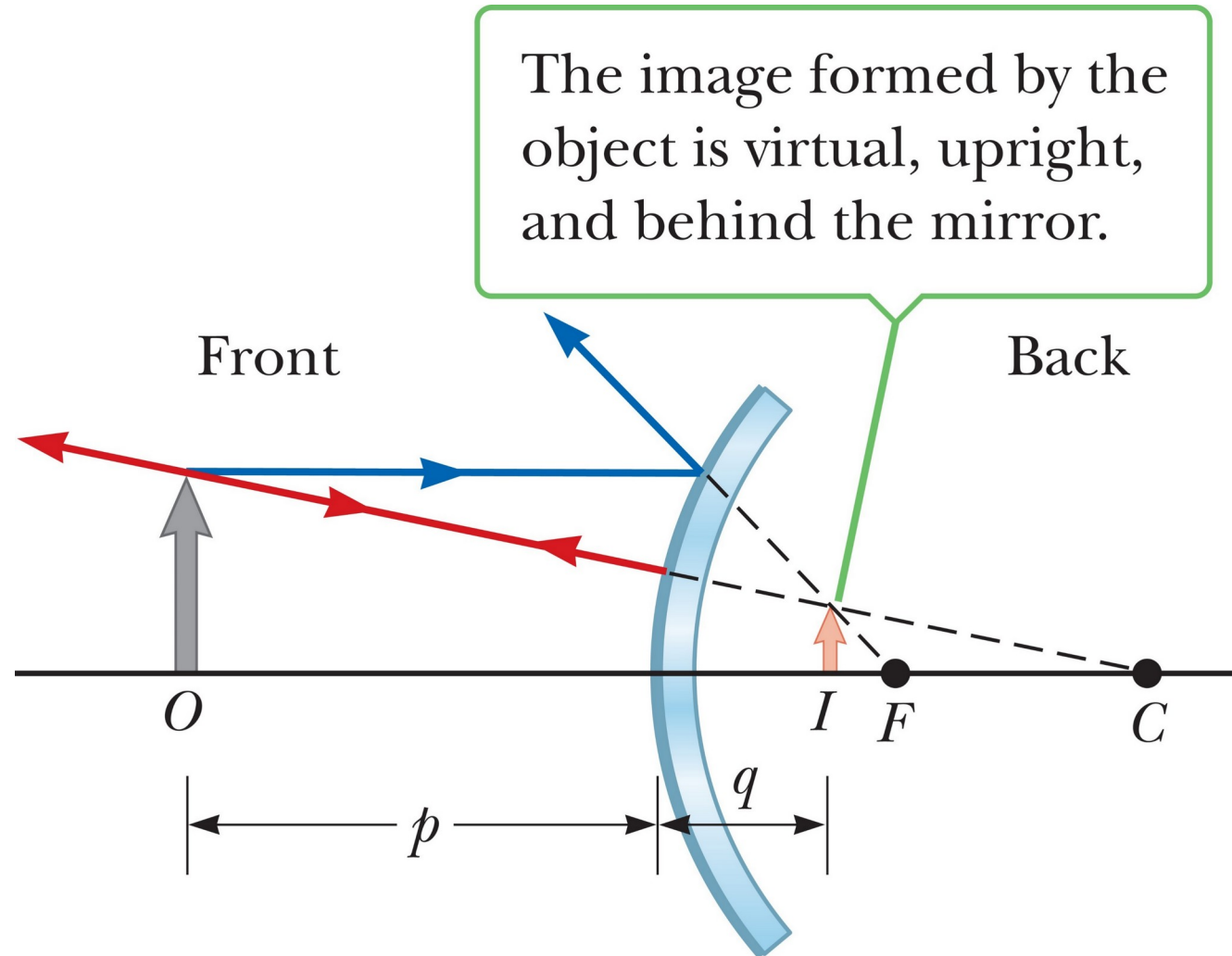
$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$$

For p approaching infinity, q approaches $R/2 = f$, the focal length of the mirror.



Henry Leap and Jim Lehman

Convex Mirrors and Sign Conventions (1 of 2)



Convex Mirrors and Sign Conventions (2 of 2)

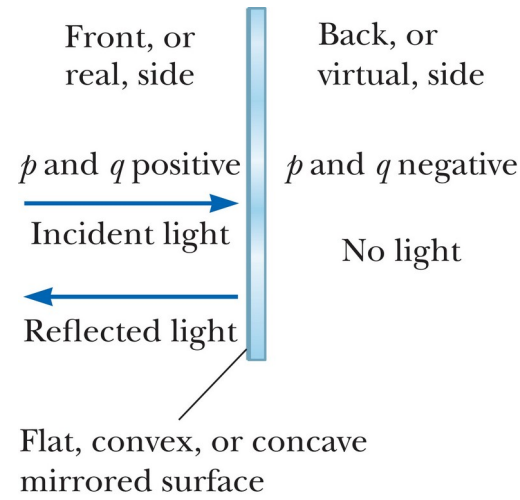
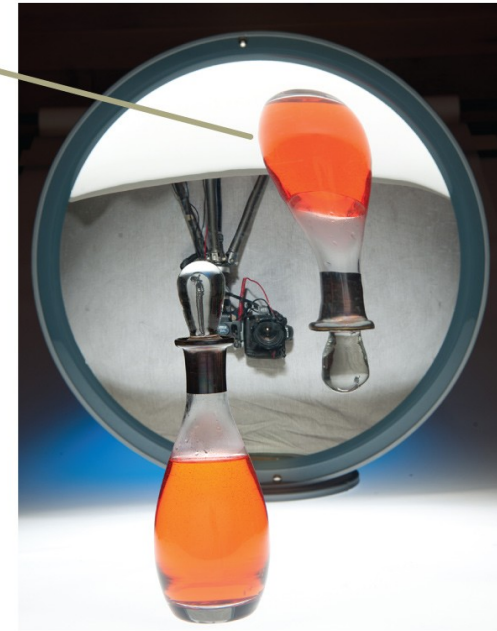
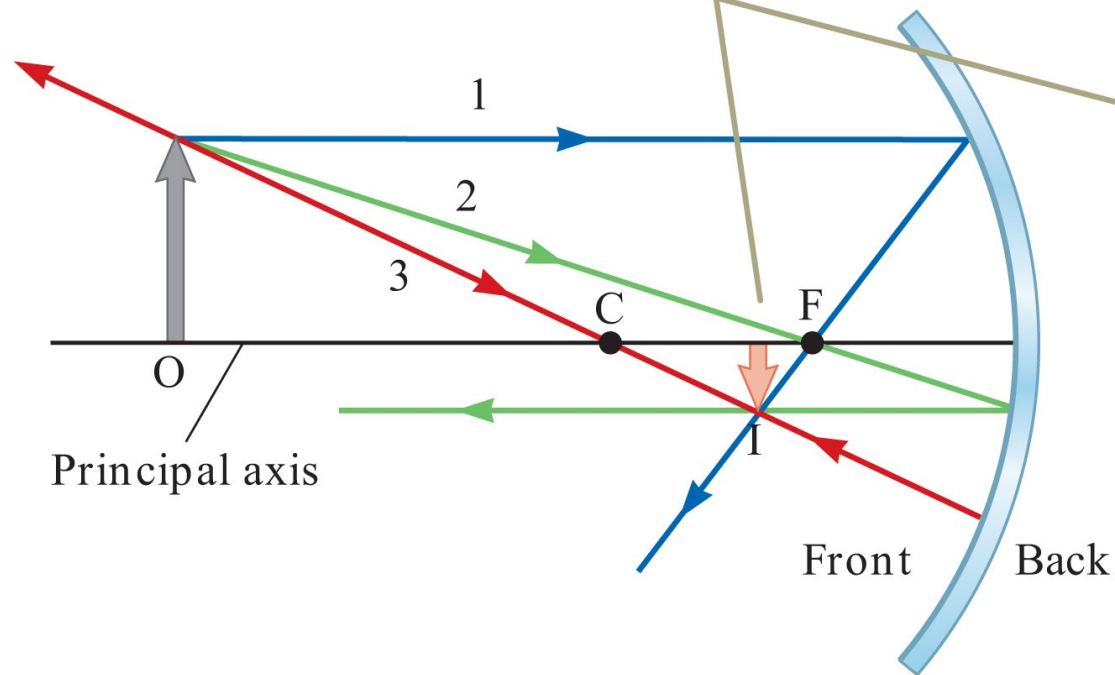


Table 23.1 Sign Conventions for Mirrors

Quantity	Symbol	In Front	In Back	Upright Image	Inverted Image
Object location	p	+	−		
Image location	q	+	−		
Focal length	f	+	−		
Image height	h'			+	−
Magnification	M			+	−

Ray Diagrams for Mirrors (1 of 4)

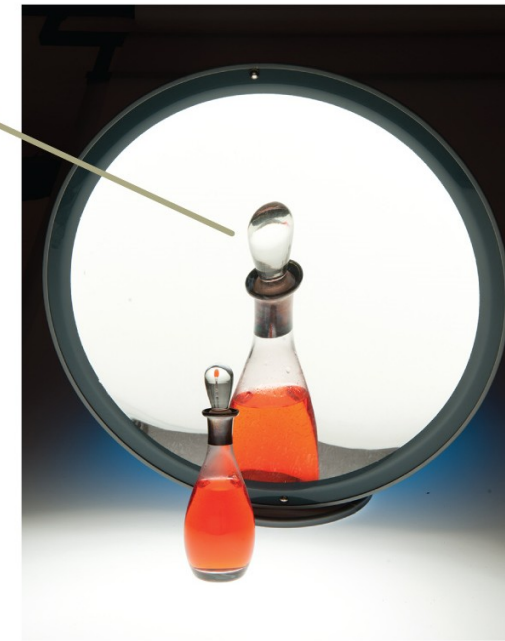
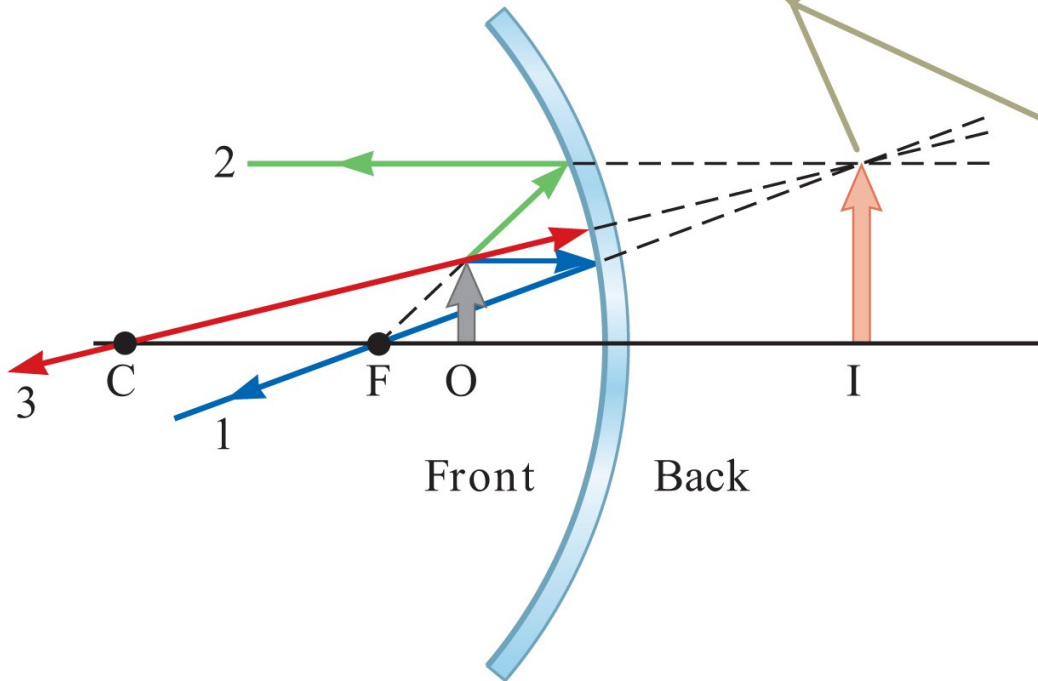
When the object is located so that the center of curvature lies between the object and a concave mirror surface, the image is real, inverted, and reduced in size.



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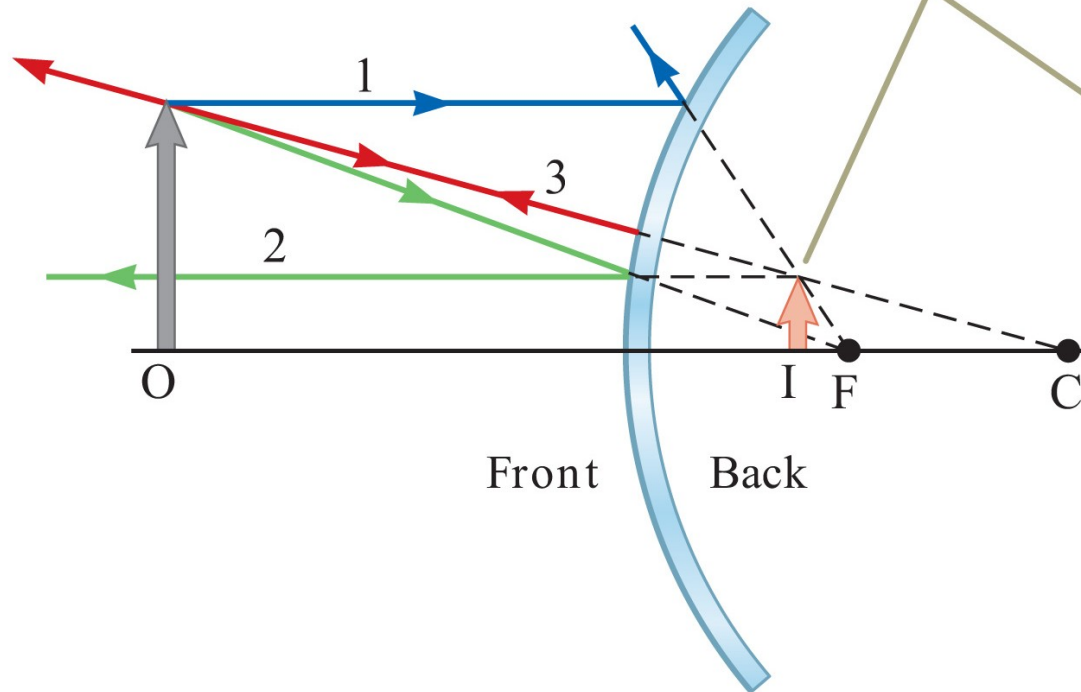
Ray Diagrams for Mirrors (2 of 4)

When the object is located between the focal point and a concave mirror surface, the image is virtual, upright, and enlarged.



Ray Diagrams for Mirrors (3 of 4)

When the object is in front of a convex mirror, the image is virtual, upright, and reduced in size.

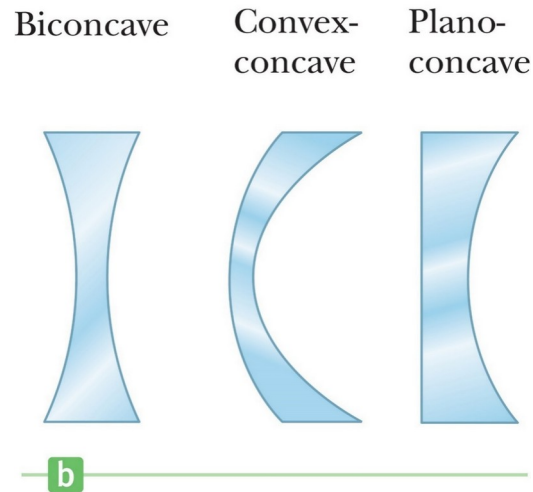
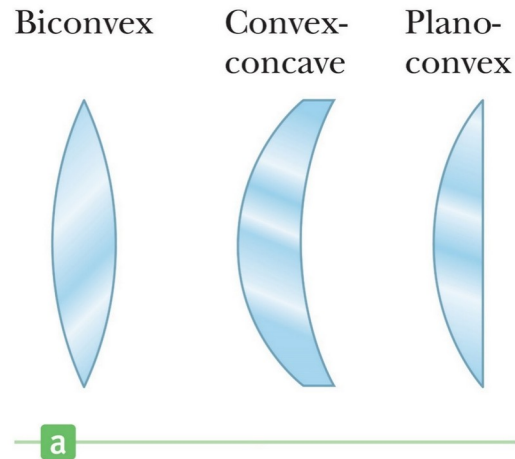


Ray Diagrams for Mirrors (4 of 4)

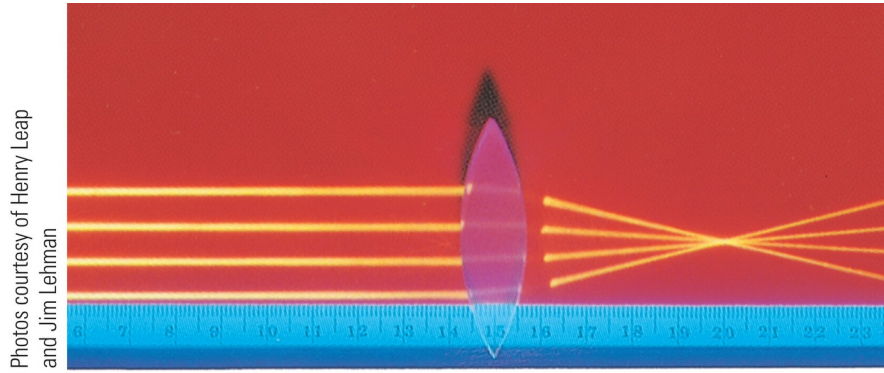


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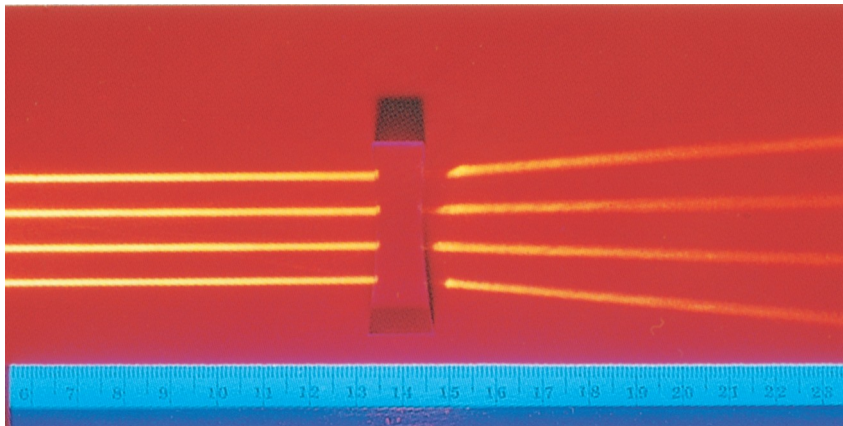
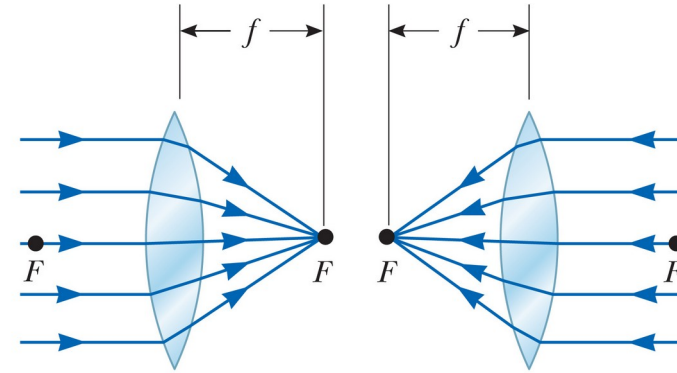
Thin Lenses (1 of 5)



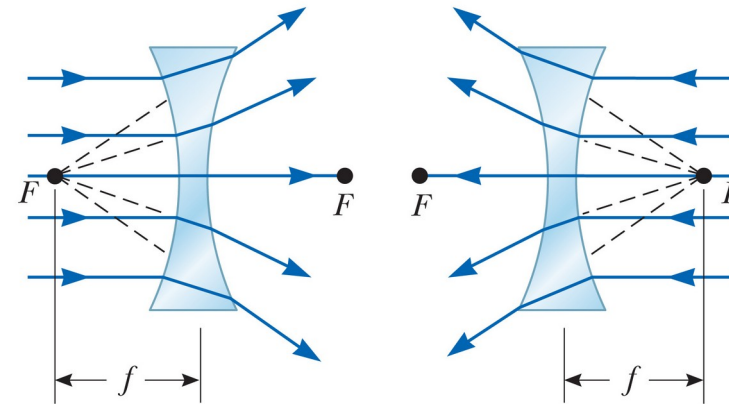
Thin Lenses (2 of 5)



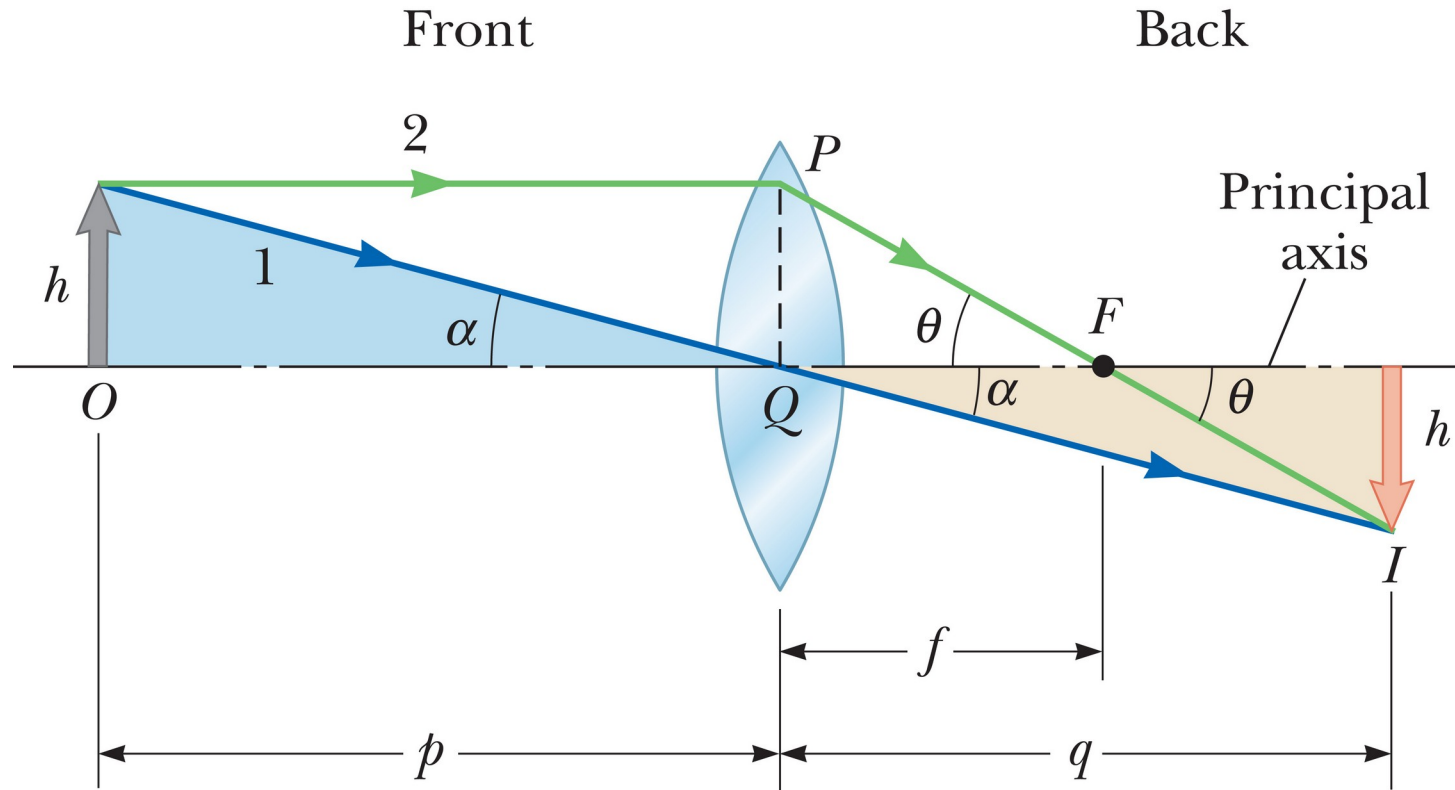
a



b



Thin Lenses (3 of 5)



$$\tan \alpha = \frac{h}{p} \quad \text{and} \quad \tan \alpha = -\frac{h'}{q}$$

$$M = \frac{h'}{h} = -\frac{q}{p}$$

$$\tan \theta = \frac{PQ}{f} = -\frac{h'}{q-f}$$

$$\frac{h}{f} = -\frac{h'}{q-f} \rightarrow \frac{h'}{h} = -\frac{q-f}{f}$$

Thin Lenses (4 of 5)

$$\frac{q}{p} = \frac{q - f}{f} \rightarrow \frac{1}{p} + \frac{1}{q} = \frac{1}{f}$$

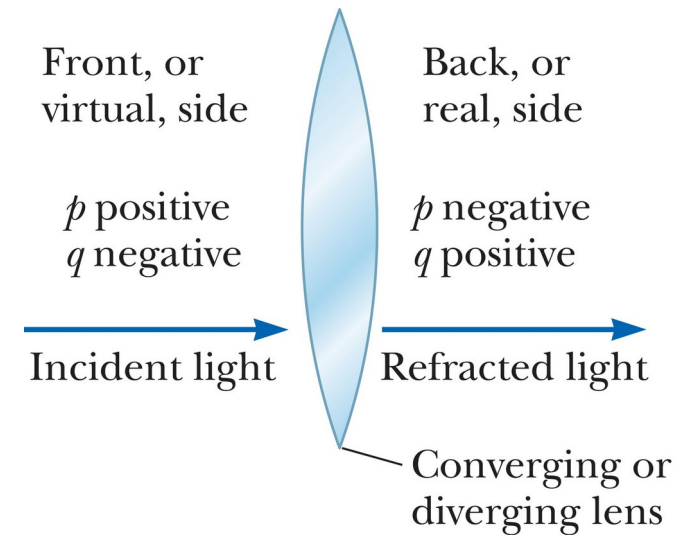


Table 23.3 Sign Conventions for Thin Lenses

Quantity	Symbol	In Front	In Back	Convergent	Divergent
Object location	p	+	−		
Image location	q	−	+		
Lens radii	R_1, R_2	−	+		
Focal length	f			+	−

Thin Lenses (5 of 5)

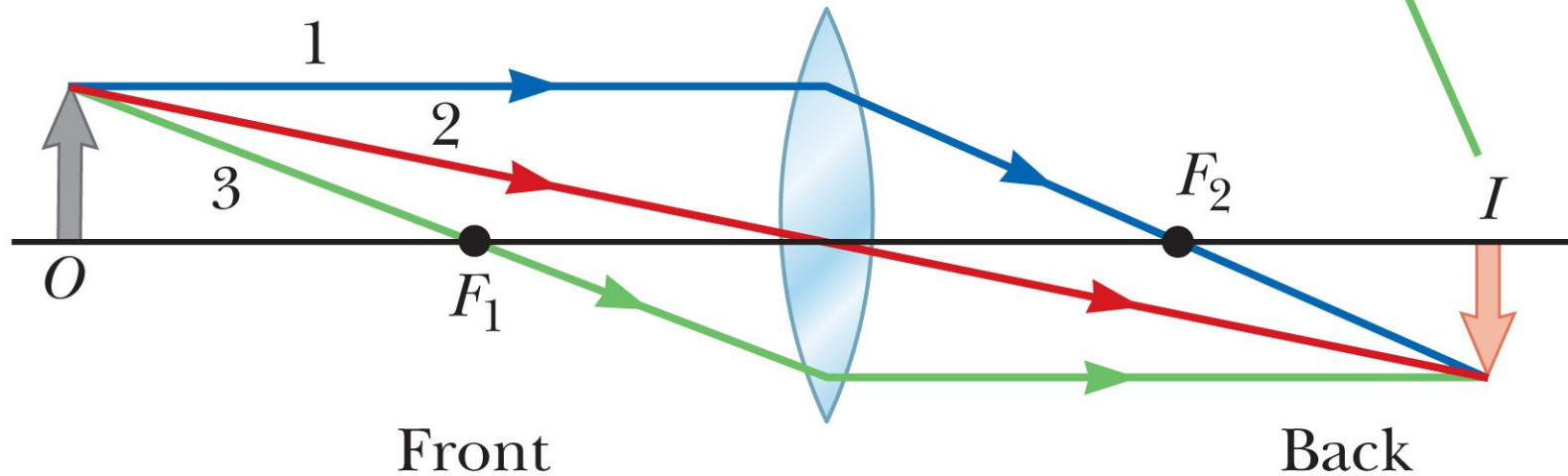
$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Table 23.3 Sign Conventions for Thin Lenses

Quantity	Symbol	In Front	In Back	Convergent	Divergent
Object Location	p	+	−		
Image Location	q	−	+		
Lens radii	R_1, R_2	−	+		
Focal length	f			+	−

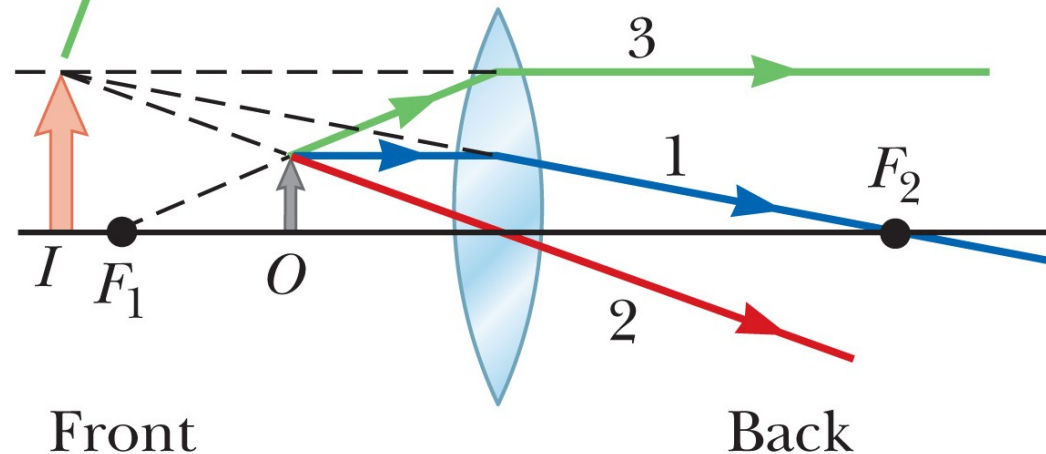
Ray Diagrams for Thin Lenses (1 of 3)

When the object is in front of and outside the focal point of a converging lens, the image is real, inverted, and on the back side of the lens.



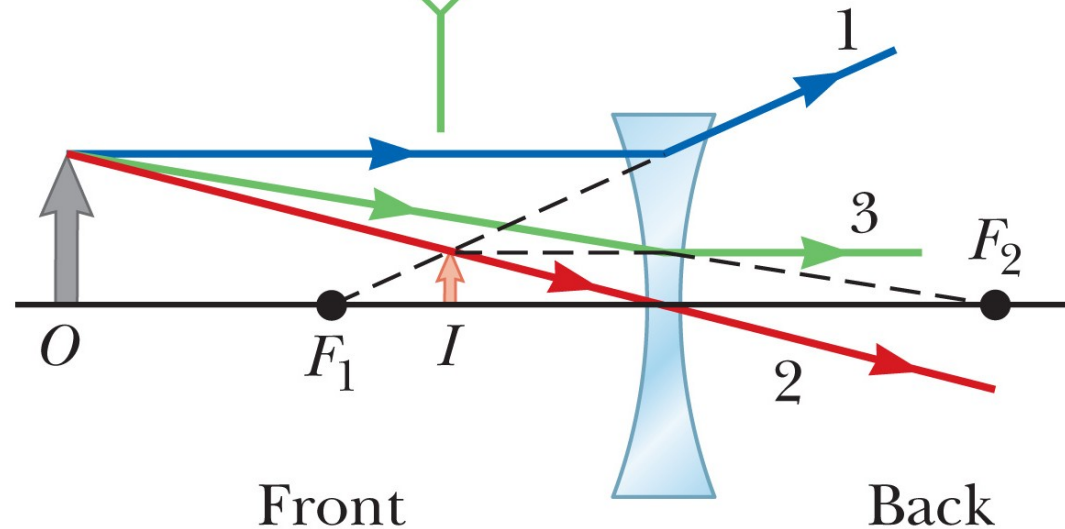
Ray Diagrams for Thin Lenses (2 of 3)

When the object is between the focal point and a converging lens, the image is virtual, upright, larger than the object, and on the front side of the lens.



Ray Diagrams for Thin Lenses (3 of 3)

When an object is anywhere in front of a diverging lens, the image is virtual, upright, smaller than the object, and on the front side of the lens.



Think – Pair – Share 2

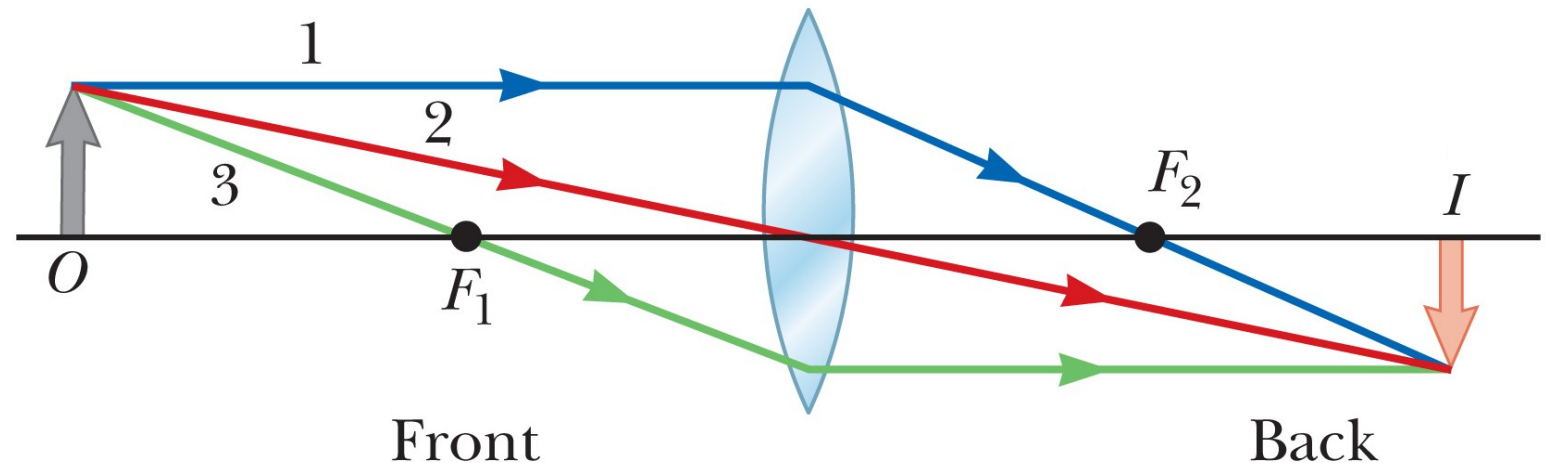
A clear plastic sandwich bag filled with water can act as a crude converging lens in air. If the bag is filled with air and placed under water, the effective lens is

1. converging
2. diverging

Think – Pair – Share 3

In the figure below the blue object arrow is replaced by one that is much taller than the lens. How many rays from the object will strike the lens?

1. 3
2. 2
3. 1
4. none
5. an infinite number



Think – Pair – Share 4

True or False? An object is placed to the left of a converging lens. The image is always to the right of the lens.

1. True
2. False

Think – Pair – Share 5

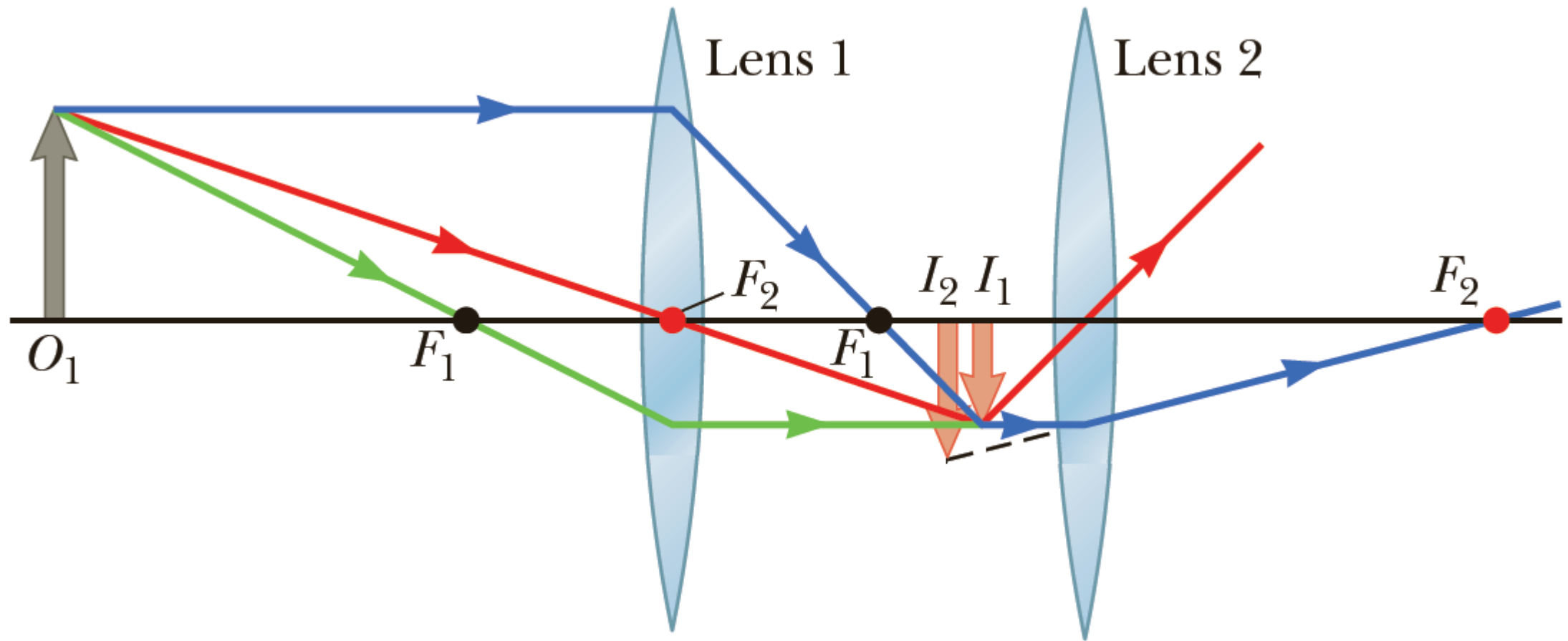
True or False: An object is placed to the left of a converging lens. The image can be upright or inverted.

1. True
2. False

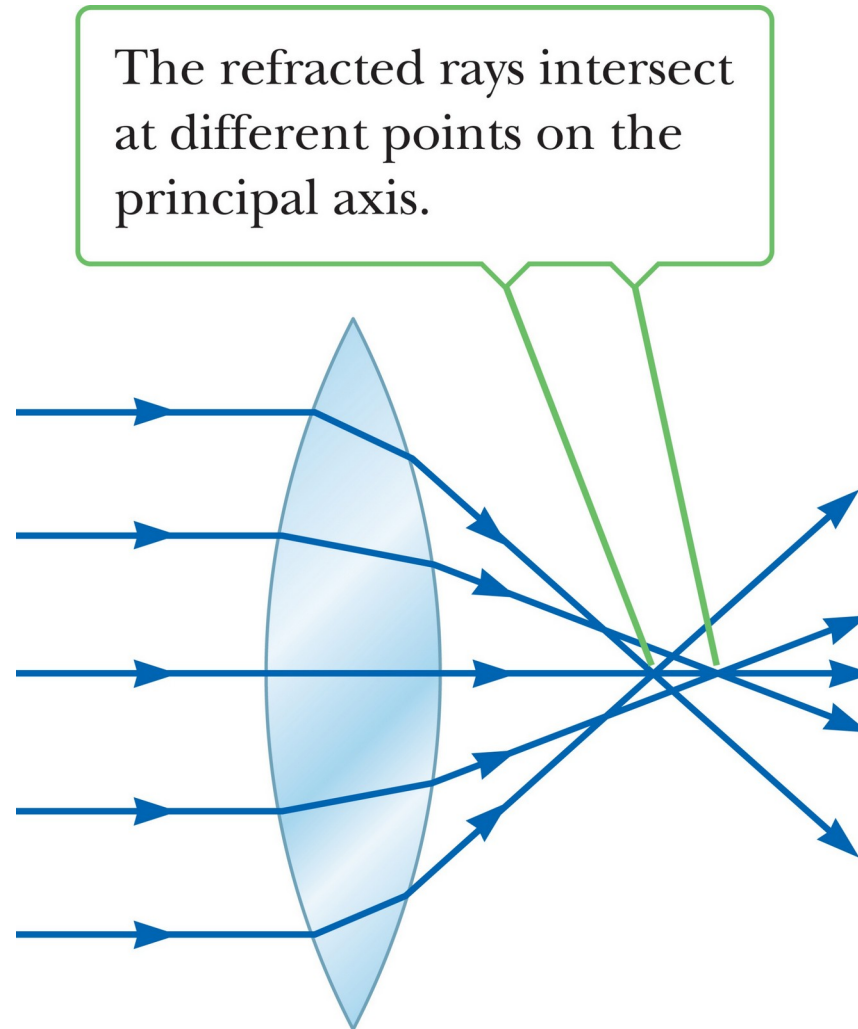
Think – Pair – Share 6

True or False? An object is placed to the left of a converging lens. The image is always smaller or the same size as the object.

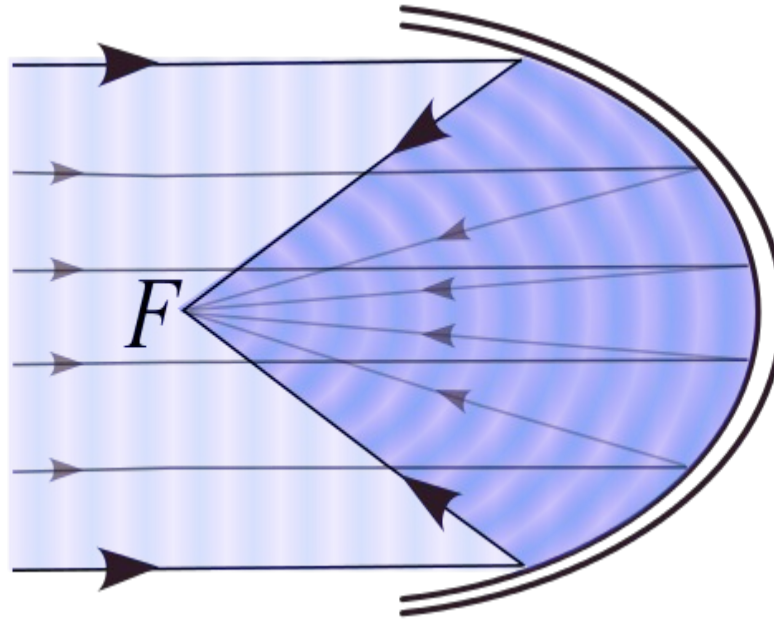
Combinations of Thin Lenses



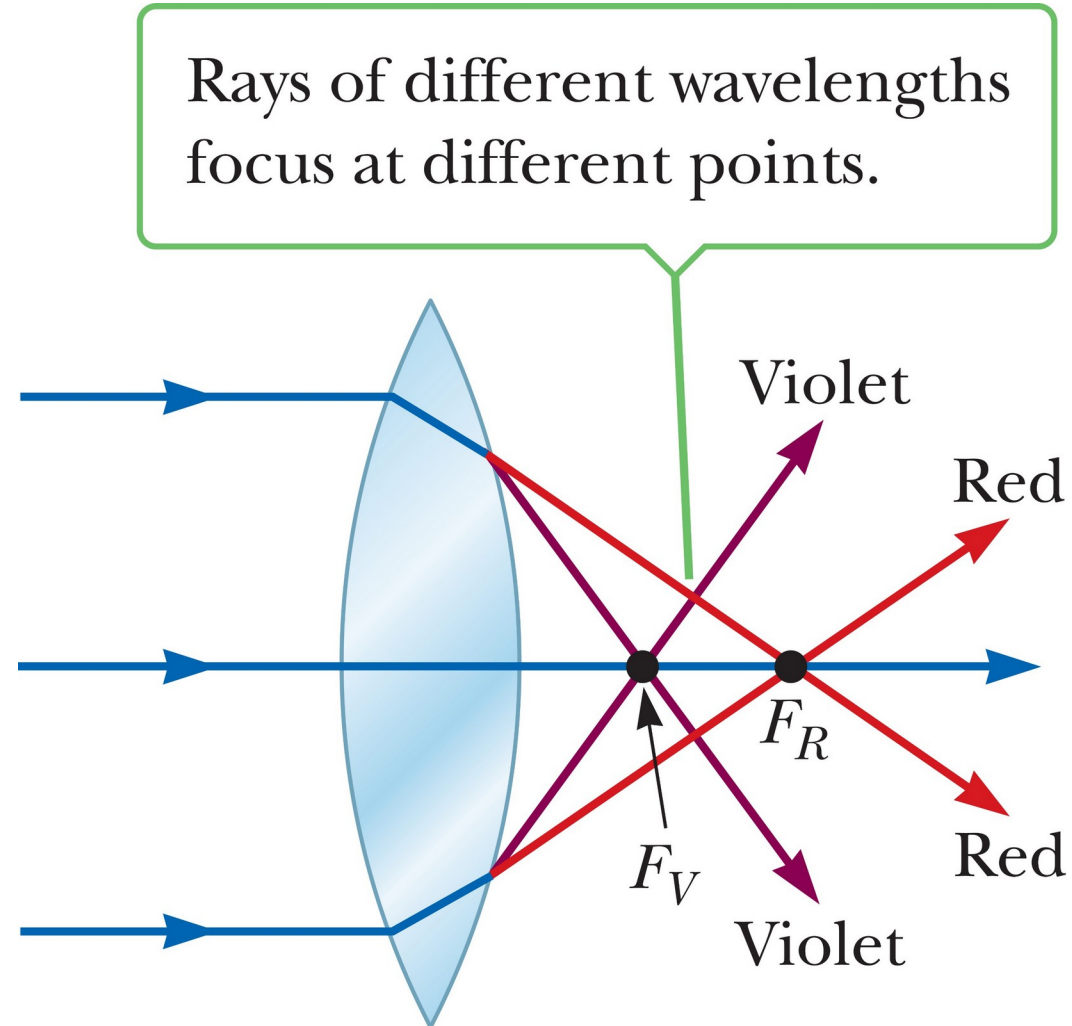
Spherical Aberration (1 of 2)



Spherical Aberration (2 of 2)

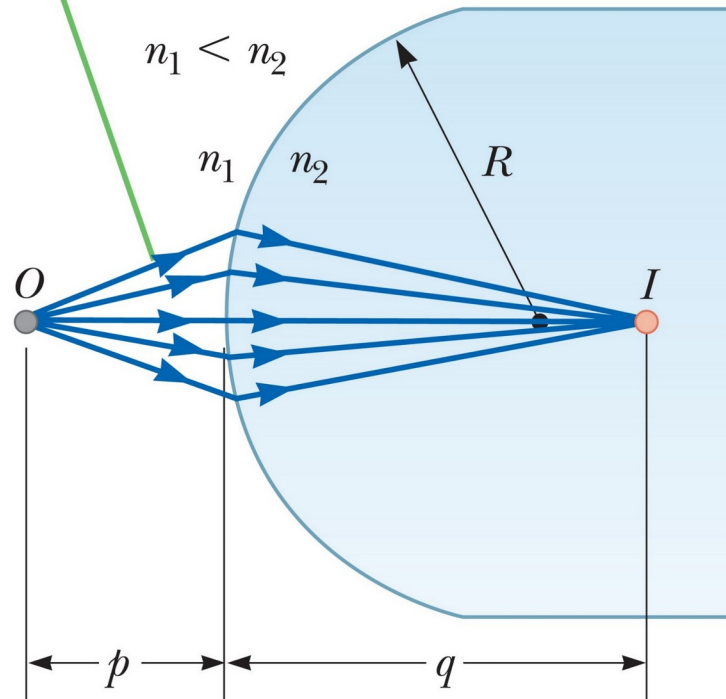


Chromatic Aberration



Images Formed by Refraction (1 of 2)

Rays making small angles with the principal axis diverge from a point object at O and pass through the image point I .



$$\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$$

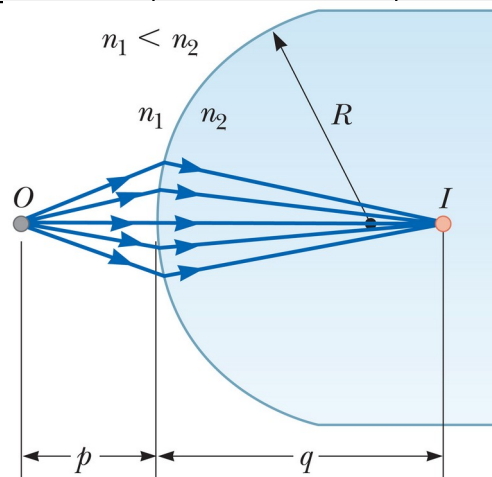
$$M = \frac{h'}{h} = -\frac{n_1 q}{n_2 p}$$

$$\frac{n_1}{p} = -\frac{n_2}{q} \rightarrow q = -\frac{n_2}{n_1}p$$

Images Formed by Refraction (2 of 2)

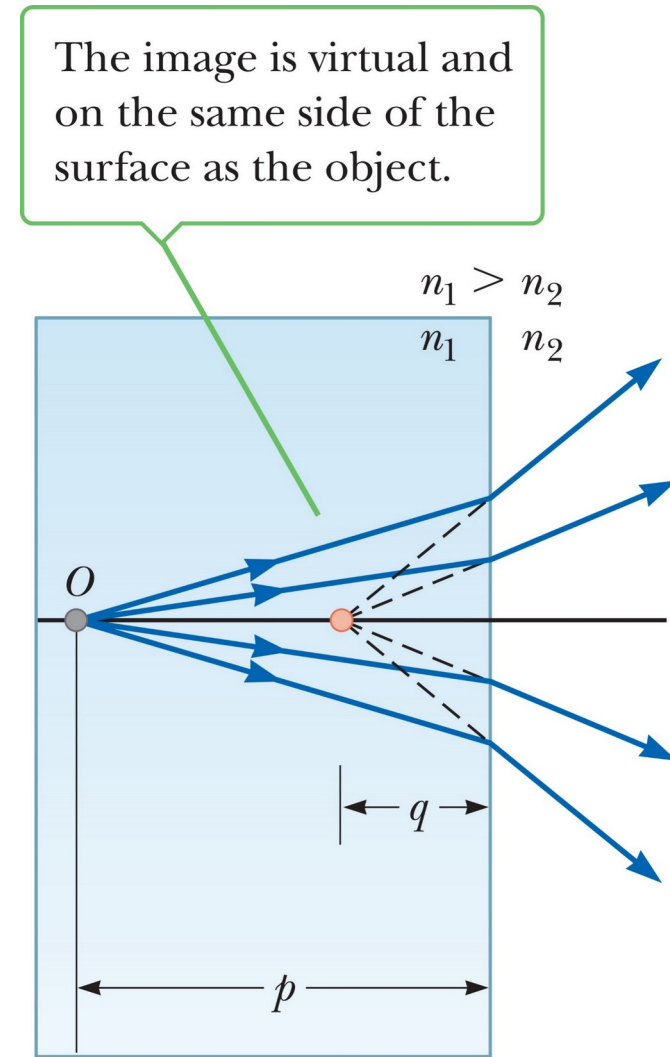
Table 23.2 Sign Conventions for Refracting Surfaces

Quantity	Symbol	In Front	In Back	Upright Image	Inverted Image
Object location	p	+	−		
Image location	q	−	+		
Radius	R	−	+		
Image height	h'			+	−



Flat Refracting Surfaces

$$q = -\frac{n_2}{n_1}p$$



Think – Pair – Share 7

A person spearfishing from a boat sees a fish located 3 m from the boat at an apparent depth of 1 m. To spear the fish, the person should aim

1. at the image of the fish.
2. above the image of the fish.
3. below the image of the fish.

Think – Pair – Share 8

True or False? The image of an object placed in front of a concave mirror is always upright.

1. True
2. False

Think – Pair – Share 9

True or False? The height of the image of an object placed in front of a concave mirror must be smaller than or equal to the height of the object.

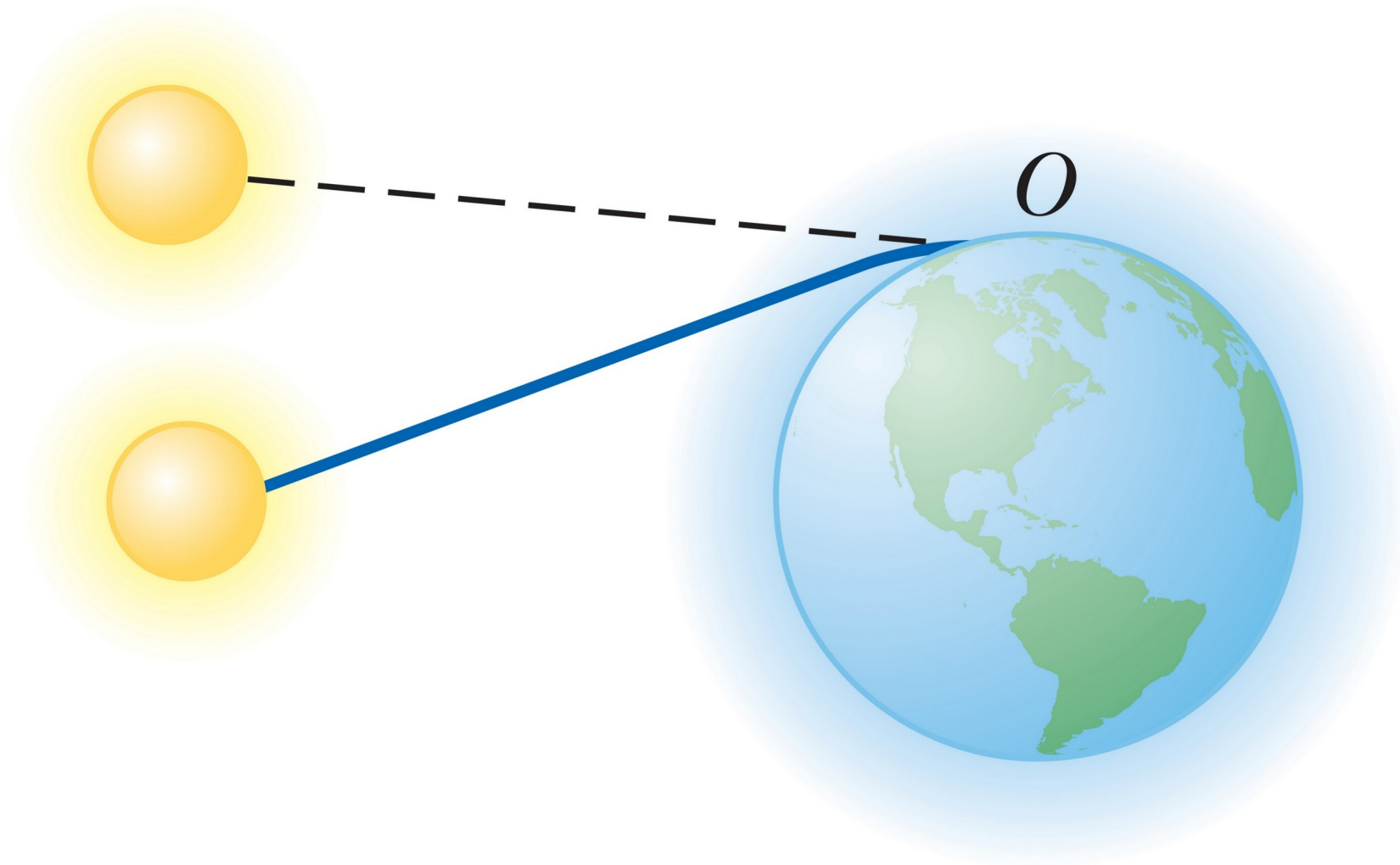
1. True
2. False

Think – Pair – Share 10

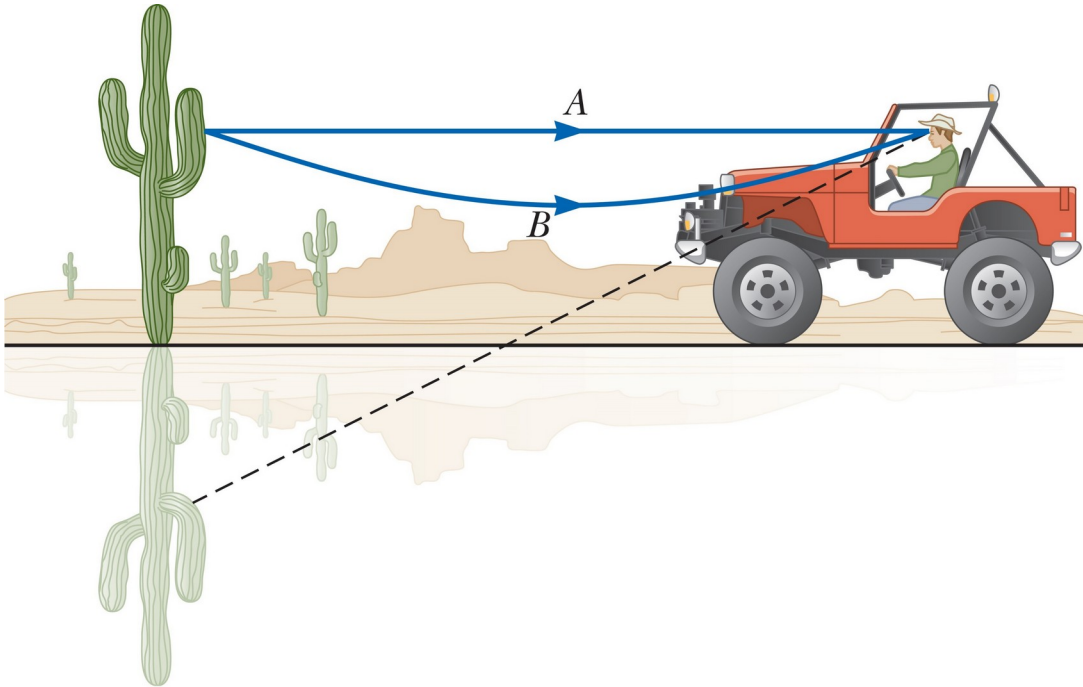
True or False? The image of an object placed in front of a convex mirror is always upright and smaller than the object.

1. True
2. False

Atmospheric Refraction (1 of 2)



Atmospheric Refraction (2 of 2)



a



b